THE NOVA RESEARCH CURRICULUM

Mathematics

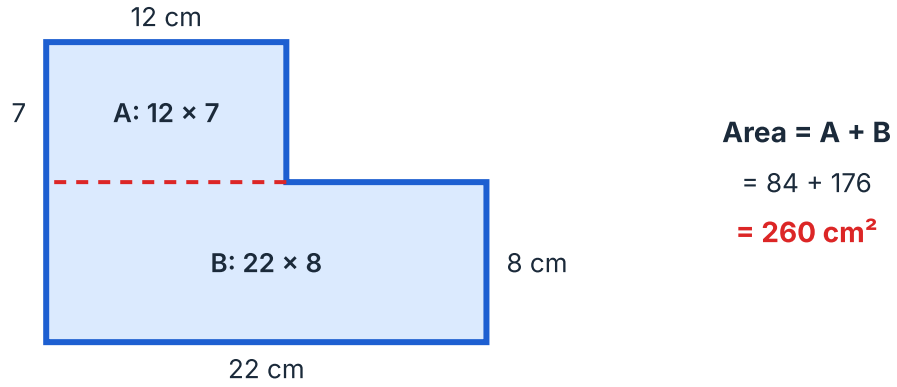
Grade 6 · Term 3

Student edition

Outcomes: MA-6.5 perimeter, area and volume; MA-6.6 data handling and probability.

Lesson notes

- **MA-6.5 Perimeter and area (Weeks 1–4):** perimeter = distance round the edge. Area of rectangle = length × width; triangle = ½ × base × height; composite shapes by splitting into rectangles. Units cm², m². Measure the classroom floor.



Split a composite shape into rectangles and add their areas.

- **MA-6.5 Volume (Weeks 5–8):** volume of a cuboid = length × width × height, in cm³; 1 litre = 1,000 cm³. Build cuboids from unit cubes.

  ▶ **Try it on NovaSchool:** Stacking cubes: volume of a cuboid — open **Simulations** on your dashboard.

- **MA-6.6 Data and probability (Weeks 9–12):** collect data, tally, bar chart and line graph; mean, median, mode and range. Probability scale from 0 (impossible) to 1 (certain); probability of an outcome = favourable ÷ total (a coin, a dice).
- **Watch out:** mixing perimeter and area units; forgetting ½ for triangles; not ordering data before finding the median.

Workbook pages

Page	Task
1–2	Perimeter and area of 10 shapes, including composites.
3	Volume of 6 cuboids; litres conversion.
4–5	Class survey: tally, bar chart, mean, median, mode, range.
6	Coin and dice experiment; compare with theory.

Question bank

#	Outcome	Question	Your answer
1	MA-6.5	Area of a rectangle 8 cm by 5 cm?	13 cm² / 26 cm² / 40 cm² / 40 cm
2	MA-6.5	Perimeter of a square with side 6 cm?	12 cm / 24 cm / 36 cm / 36 cm²
3	MA-6.5	Area of a triangle with base 10 cm and height 4 cm?	40 cm² / 20 cm² / 14 cm² / 7 cm²
4	MA-6.5	Volume of a box 5 cm × 4 cm × 3 cm?	12 cm³ / 47 cm³ / 60 cm³ / 60 cm²
5	MA-6.5	1 litre = 1,000 cm³. True or false.	True / False
6	MA-6.6	Mean of 4, 6, 8, 10?	6 / 7 / 8 / 28
7	MA-6.6	Median of 3, 9, 1, 7, 5?	9 / 1 / 5 / 7
8	MA-6.6	Mode of 2, 3, 3, 5, 7?	2 / 3 / 5 / 7
9	MA-6.6	Probability of rolling a 6 on a fair dice?	1/6 / 1/2 / 6 / 0
10	MA-6.6	The sun rising tomorrow has probability close to …	0 / 1/2 / 1 / 2